

SGA 6 – Exposé XIV

Section 4.7 (continuation) through the end

Source coordinates: current-rescribe idx693–idx702; printed pages 680–689; source-PDF pages 683–692. The opening continues the sentence at the exact idx692 seam. The source scan controls this terminal range; later French and inherited English files are locator and comparison witnesses only.

4.7 (continuation). only if G is “without torsion”, i.e. if and only if G is isomorphic to a product of groups $\mathrm{Sl}(n)$ and $\mathrm{Sp}(m)$. Thus, in this case and only in this case, the homomorphism

$$\varphi : A(G) \longrightarrow \mathrm{Gr}_{\mathrm{top}}^{\bullet}(G)$$

is an isomorphism. Consequently, if G has torsion, the underlying scheme provides an example of a non-bijective φ .

4.8. One likewise finds an example in which the homomorphism γ of (4.1)

$$\gamma : A(G) \longrightarrow H^{2*}(G, \mathbb{Z}_{\ell}(*)) \tag{4.9}$$

does not factor through $\mathrm{Gr}_{\mathrm{top}}^{\bullet}(G)$, or, what amounts to the same thing here, an example in which this homomorphism is nonzero in degrees > 0 . Restricting (as one may, thanks to suitable specialization theorems) to the case $k = \mathbb{C}$, one is reduced to studying the homomorphism

$$A(G) \longrightarrow H^*(G(\mathbb{C}), \mathbb{Z}), \tag{4.10}$$

which is moreover obtained, by passage to the quotient, from the canonical homomorphism

$$A(G/B) \simeq H^*(G/B(\mathbb{C}), \mathbb{Z}) \longrightarrow H^*(G(\mathbb{C}), \mathbb{Z}).$$

It is tempting to conjecture that (4.10) is in fact injective, since it is also known that the torsion primes for $G(\mathbb{C})$ are exactly those dividing the order of the finite group

$$A^+(G) = \coprod_{i>0} A^i(G).$$

If this were so, whenever G had torsion, (4.10) would not factor through φ ; hence, for a suitable ℓ (more precisely, for ℓ dividing the order of $A^+(G)$, i.e. for ℓ a torsion prime of G), the same would hold for (4.9). We can at least assert that, for the group $G = G_2$ over $\mathbb{C}$, (4.10) is nonzero in degree > 0 : more precisely, there exists an element of $A^3(G)$ whose image in $H^6(G(\mathbb{C}), \mathbb{Z})$ has order 2. It follows that (4.9) does not factor through φ when $\ell = 2$. The fact used about G_2 , which amounts to saying that there exists in $H^6(G/B(\mathbb{C}), \mathbb{Z})$ an element whose image in $H^6(G(\mathbb{C}), \mathbb{Z})$ is nonzero, is found in [3], via the familiar dictionary between complex semisimple algebraic groups and compact Lie groups; it was kindly communicated to me by J.-P. Serre.

5. Relations between $\mathrm{Gr}^{\bullet}(X)$ and $H^{2*}(X, \mathbb{Z}_{\ell}(*))$

5.1. The example given in 4.8 does not exclude the possibility that, for an arbitrary scheme X , one might find a ring homomorphism

$$g' : \mathrm{Gr}^{\bullet}(X) \longrightarrow H^{2*}(X, \mathbb{Z}_{\ell}(*)),$$

such that, for every locally free module E on X , g^i sends the Chern class $c_i(E)$ of Exp. V to the cohomological Chern class of SGA 5 VII. Of course, ℓ here denotes a prime distinct from the residual characteristics of X . Since the ring $\text{Gr}^\bullet(X)$ is generated by the $c^i(E)$, one sees in any case that g' is uniquely determined by the requirement that it be compatible with Chern classes.

5.2. Using the Chern homomorphism

$$K^\bullet(X) \longrightarrow \text{Ch}(H^{2*}(X, \mathbb{Z}_\ell(*))),$$

one obtains, by passing to the associated gradeds respectively for the filtration on the first member and the filtration by degree on the second, a homomorphism of graded groups (but not of rings!)

$$\psi : \text{Gr}^\bullet(X) \longrightarrow H^{2*}(X, \mathbb{Z}_\ell(*)),$$

which satisfies

$$\psi(c_i(E)) = (-1)^{i-1}(i-1)! c_i^\ell(E),$$

where $c_i^\ell(E)$ denotes the ℓ -adic Chern class. Dividing the components ψ^i by the indicated factors therefore gives a ring homomorphism

$$g_{\mathbb{Q}} : \text{Gr}^\bullet(X) \longrightarrow H^{2*}(X, \mathbb{Q}_\ell(*)) = H^{2*}(X, \mathbb{Z}_\ell(*)) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell,$$

sending Chern classes to Chern classes, and such that the image of the first member is contained in that of $H^2(X, \mathbb{Z}_\ell(*))$. Consequently, the question of the existence of g' has an affirmative answer provided $H^{2*}(X, \mathbb{Z}_\ell(*))$ is replaced by its quotient by its torsion subgroup. The first doubtful case for defining the components g^i of the hypothetical homomorphism g is that of g^3 ; a priori one succeeds only in defining $2g^3 = \psi^3$. The answer to the question of the existence of g^3 would be negative if, for example, one could find three invertible sheaves L_1, L_2, L_3 on X satisfying

$$(L_1 - 1)(L_2 - 1)(L_3 - 1) = 0 \quad \text{in } K^\bullet(X),$$

but such that

$$c_1(L_1)c_1(L_2)c_1(L_3) \neq 0 \quad \text{in } H^6(X, \mu_2^{\otimes 3}).$$

5.3. When X is smooth and quasi-projective over a field k , one may similarly ask whether there exists a ring homomorphism

$$g : \text{Gr}^\bullet(X) \longrightarrow A(X)$$

taking values in the Chow ring $A(X)$ and sending Chern classes to Chern classes. The preceding observations may be repeated in this connection; they show in particular how to construct $(-1)^{i-1}(i-1)!g^i$ directly when g^i itself is unavailable.

5.4. Let us note, to conclude this section, that essentially the same question arises if, instead of a scheme X , one considers a compact topological space (say) X . One can still consider the associated graded $\text{Gr}^\bullet(X)$ of $K^\bullet(X)$ endowed with its λ -filtration, and seek a ring homomorphism

$$\text{Gr}^\bullet(X) \longrightarrow H^{2*}(X, \mathbb{Z})$$

sending Chern classes to Chern classes.

6. Cohomological Riemann–Roch theorem and Gysin homomorphism

6.1. For simplicity, we shall confine ourselves here to schemes X admitting an ample invertible module, so that there is no need to distinguish between the naive $K^\bullet(X)$ and the other one. Let us begin with the Riemann–Roch formula “with denominators”, for a proper complete-intersection morphism $f : X \rightarrow Y$. For a prime ℓ prime to the residual characteristics of Y , one can define the ℓ -adic Chern classes of a locally free module E ,

$$c_i(E) \in H^{2i}(X, \mathbb{Z}_\ell(i))$$

(SGA 5 VII), corresponding to a homomorphism

$$K^\bullet(X) \rightarrow \text{Ch}(H^{2*}(X, \mathbb{Z}_\ell(*))). \quad (6.1)$$

The question therefore arises of giving a cohomological version of the Riemann–Roch formula. The only ingredient missing in order to give meaning to the formula is a Gysin homomorphism

$$H^n(X, \mathbb{Q}_\ell(i)) \rightarrow H^{n-2d}(Y, \mathbb{Q}_\ell(i-d)), \quad (6.2)$$

where d is the virtual relative dimension of f . Assuming this homomorphism to be defined, and taking into account the Riemann–Roch formula in the form of Exp. VIII, with values in $\text{Gr}^\bullet(Y)_\mathbb{Q}$, one sees that the validity of the ℓ -adic Riemann–Roch formula for the morphism f is exactly equivalent to the commutativity of the square

$$\begin{array}{ccc} \text{Gr}^\bullet(X)_\mathbb{Q} & \longrightarrow & H^{2*}(X, \mathbb{Q}_\ell(*)) \\ \downarrow & & \downarrow \\ \text{Gr}^\bullet(Y)_\mathbb{Q} & \longrightarrow & H^{2*}(Y, \mathbb{Q}_\ell(*)), \end{array} \quad (6.3)$$

where the horizontal arrows are obtained from (6.1) by passage to the associated gradeds and division, so as to send Chern classes to Chern classes, while the vertical arrows are respectively the arrow of Exp. VIII and the Gysin homomorphism in cohomology (assumed given).

6.2. Note that the Gysin homomorphism is defined in all cases if Y is smooth over a field k (SGA 5 IV), using the theorem of relative cohomological purity over k . The same construction applies when Y is excellent, regular, and of characteristic zero, thanks to the theorem of absolute cohomological purity of SGA 4 XIX; and it will automatically apply to every excellent regular Y , without restriction on the characteristic, once the theorem of absolute cohomological purity has been proved in full generality. Finally, the same construction applies, thanks to the theorem of relative cohomological purity (SGA 4 XVI 3.7), when X and Y are both smooth over the same scheme S and f is an S -morphism. It is natural to conjecture that, in each of these cases in which the Gysin homomorphism is defined, diagram (6.3) is indeed commutative. The only case in which this assertion is currently verified is that in which X and Y are both smooth over a field k ; it then follows from the commutativity of the analogous diagram

$$\begin{array}{ccc} A(X) & \longrightarrow & H^{2*}(X, \mathbb{Z}_\ell(*)) \\ \downarrow & & \downarrow \\ A(Y) & \longrightarrow & H^{2*}(Y, \mathbb{Z}_\ell(*)), \end{array} \quad (6.4)$$

involving the rings $A(X)$ and $A(Y)$ of cycle classes modulo rational equivalence, whose commutativity was established in SGA 5 IV.

6.3. Note that in each of the cases considered above in which one succeeds in defining a Gysin homomorphism, one in fact obtains a construction for cohomology with coefficients in $\mathbb{Z}_\ell(*)$, and not only in $\mathbb{Q}_\ell(*)$. This makes it possible, when f is an *immersion*, to pose the question of the validity of the Riemann–Roch formula “without denominators”, as explained in no. 3. When X and Y are both smooth over a field k , the commutativity of (6.4) shows that it would suffice to prove the formula without denominators in the Chow ring $A(Y)$; cf. (3.3).

6.4. When no regularity hypothesis is made on Y , and even if X and Y are of finite type over $\mathbb{C}$, one no longer sees a plausible definition of a cohomological Gysin homomorphism (6.2), even for a regular immersion of codimension 1. Can one nevertheless find one with reasonable properties, in particular making square (6.3) commutative? The question is evidently connected with that of defining, for X a complete intersection of codimension i in Y , “its cohomology class”

$$\gamma(X) \in H^{2i}(Y, \mathbb{Z}_\ell(i)),$$

in such a way as to turn intersections (without excess components) into cup products. This latter question has an affirmative solution at least modulo torsion, i.e. in $H^{2*}(Y, \mathbb{Q}_\ell(*))$, by defining

$$\gamma(X) = \frac{1}{(-1)^{i-1}(i-1)!} c_i(\mathcal{O}_X).$$

But I do not know whether $c_i(\mathcal{O}_X)$ is divisible by $(i-1)!$ in $H^{2i}(Y, \mathbb{Z}_\ell(i))$. In particular, if Y is, say, a projective scheme of dimension 3 over the field of complex numbers and X is reduced to a (singular) point of Y , is $c_3(\mathcal{O}_X)$ divisible by 2 in $H^6(Y, \mathbb{Z}_2(3))$, and in particular is the Chern class zero in $H^6(Y, \mathbb{Z}/2\mathbb{Z})$? Let us also mention, along the same lines, the following question: let Y be a projective (singular) scheme of dimension 4, and X a closed subscheme of dimension 1 which is globally an intersection of three positive divisors D_1, D_2, D_3 . Consider the product of their Chern classes modulo 2,

$$c_1(D_1)c_1(D_2)c_1(D_3) \in H^6(Y, \mathbb{Z}/2\mathbb{Z});$$

is this element of H^6 independent of the particular choice of the D_i whose intersection is X ?

7. Chern classes of perfect complexes

7.1. When one poses the question of developing a cohomological Riemann–Roch formula, with or without denominators, outside the rather artificial hypothesis that an ample invertible module exists on the schemes under consideration, one is necessarily faced with the question of defining, for a perfect complex of modules $L^\bullet$ on a scheme X , the Chern classes

$$c_i(L^\bullet) \in H^{2i}(X, \mathbb{Z}_\ell(i)),$$

generalizing the Chern classes of locally free modules, SGA 5 VII. The same question arises, moreover, for general locally ringed topoi (compare [10]), as well as for perfect complexes on paracompact spaces ringed by continuous complex-valued functions, working then with ordinary integral cohomology. For a compact space, it is known that the $K^\bullet(X)$ constructed with perfect complexes is the same as the “naive” one constructed with locally free modules (Exp. IV), and in that case one is trivially reduced to the classical definition. But already for a locally compact space the problem posed seems nontrivial, since the integral cohomology of such a space is not always the inverse limit of the cohomologies of its compact subsets.

7.2. More precisely, when Y is a closed subset of X such that $L^\bullet|_{X-Y}$ is zero (in the sense of derived categories), i.e. such that the cohomology sheaves of $L^\bullet$ are supported in Y , one would like to define Chern classes *with support in Y*

$$c_i^Y(L^\bullet) \in H_Y^{2i}(X, \mathbb{Z}_\ell(i)) \quad \text{respectively} \quad H_Y^{2i}(X, \mathbb{Z}),$$

which would recover the “ordinary” (sic) Chern classes of 7.1 by means of the canonical homomorphism

$$H_Y^*(X, -) \longrightarrow H^*(X, -).$$

Even in the case where the latter reduce to the Chern classes of locally free modules (X a scheme with an ample invertible module, or a compact space), the construction of these Chern classes “with supports” has not been carried out and will probably only be possible together with that of the general Chern classes postulated in 7.1.¹

It is probable that Chern classes with supports will have a role to play in the proof of the cohomological Riemann–Roch formula without denominators (cf. 3.5). On the other hand, they should make it possible to give an improved version of the Riemann–Roch formula (with or without denominators) by introducing families of supports into that formula.

8. Chow ring of regular schemes

The problem is to define rational equivalence of cycles and to study the resulting ring of classes (in particular its functorial properties) for a regular noetherian scheme X admitting an ample invertible module, thereby generalizing the known theory when X is smooth over a field [4]. The key technical result will again be a “moving lemma”, ensuring that a cycle can be varied within its class so as to meet a given cycle in the proper dimension.

The problem under consideration is quite natural a priori, especially since we have seen (4.5 to 4.8) that the substitutes $\text{Gr}^\bullet(X)$ and $\text{Gr}_{\text{top}}^\bullet(X)$ for the Chow ring $A(X)$ are hardly satisfactory when one does not confine oneself to a theory “modulo torsion”. When X is obtained from a smooth projective algebraic scheme Y over a field k by taking the local scheme at the vertex of a projective cone C and removing the origin, one finds (assuming that $A(X)$ is indeed the inductive limit of the Chow rings of $U - \{O\}$, where U ranges over the open neighborhoods in C of the vertex O) that $A(X)$ is canonically isomorphic to the quotient ring

$$A(Y)/\xi A(Y),$$

where $\xi \in A^1(Y)$ is induced by the chosen projective embedding of Y . This shows that knowing the Chow ring $A(X)$ (which is of a nonclassical type) is equivalent to knowing $A(Y)/\xi A(Y)$, which might be called the “primitive part” of the Chow ring $A(Y)$ (in a sense suggested by the Lefschetz–Hodge cohomological theory of nonsingular projective algebraic varieties). This example shows that the “local Chow rings” of local rings with isolated

singularities are geometric invariants that are by no means trivial and will undoubtedly have a role to play in the “arithmetic” study of these local rings.

Of course, once one has a reasonable notion of a Chow ring for a regular noetherian scheme admitting an ample invertible module, one should generalize to this case the relations established or conjectured in no. 4 between this ring and $\text{Gr}_{\text{top}}^\bullet(X)$, and formulate in terms of this ring the question of the “Riemann–Roch formula without denominators” stated in no. 3.

¹Added in October 1968. The problems raised here were solved in the framework of Hodge cohomology by L. Illusie; cf. [13].

Bibliography of Exposé XIV

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- [16](*) P. Deligne, Exposé XVIII in *Théorie des topos et cohomologie étale des schémas (SGA 6)*, by M. Artin, A. Grothendieck, and J. L. Verdier.
- [17](*) U. Mausin, *Correspondances, motifs et transformations monoïdales* (in Russian). Mat. Sbornik, t. 77 (119), no. 4, 1968.

(*) Added in February 1969.

Note for page 1

The theory of the operations Λ^i in $D^-(X)$, and of the operations λ^i in $K^\bullet(X)$, was written down by P. Deligne in 1968. He proves that $K^\bullet(X)$ thereby becomes a pre- λ -ring and, when $\mathcal{O}_X$ is a sheaf of algebras over a field, that $K^\bullet(X)$ is even a λ -ring. It is plausible that a refinement of Deligne's method would make it possible to dispense with the restrictive hypothesis on $\mathcal{O}_X$.

Notes for page 11

(*) (April 1969) D. Mumford has pointed out to us a proof of (4.6)² in the *Chow ring* (without disregarding torsion), which he obtained in a seminar at Harvard in 1959! The reader will find the proof of this beautiful result in SGA 5 VII 9. This result obviously contains the corresponding cohomological result mentioned in note (**). The cohomological result, however, is proved by J.-P. Jouanolou under more general conditions (without a quasi-projective hyperplane, and in the relative smooth case over an arbitrary base).

(**) (January 1969) J.-P. Jouanolou has just proved these formulas for every y , without disregarding torsion; cf. SGA 5 VII § 4.9. Deligne and Manin [17] had previously observed independently, and by (easy) arguments of a rather different nature, that it was enough to establish the formulas for “algebraic” y , and even for $y = 1$ in [16], in order to conclude the same relation in the general case.

Source notes. The scan consistently distinguishes asterisks in graded-total cohomology (H^{2*} and H^*) from round bullets in $K^\bullet$, $\mathrm{Gr}^\bullet$, and $\mathrm{Gr}_{\mathrm{top}}^\bullet$; those distinctions are retained here. The scan also prints $K^\bullet(X)$, rather than the prime read by later locator files, in 6.1, 7.1, and the terminal page note.

After (6.2), the scan literally says that d is the virtual relative dimension of d . The morphism in question is $f : X \rightarrow Y$ and the following sentence again names f ; the English uses the intended “of f ” under SGA6-XIV-IDX696-D-VS-F-SRCDEF-001. In the bibliography, the printed “(SGA 6)”, “U. Mausin”, and “Mat. Sborik” are retained rather than silently regularized; the final note itself cites Manin [17]. In the first note for page 11, the scan unambiguously prints “(9.6)”, although this exposé has no section 9 and the contextual result is formula (4.6). The English uses (4.6) and discloses the correction immediately under SGA6-XIV-IDX702-XREF96-VS-XREF46-SRCDEF-001.

The scan's $\coprod_{i>0} A^i(G)$, $H^2(X, \mathbb{Z}_\ell(*))$ containment, and $\mu_2^{\otimes 3}$ obstruction are retained. The admitted French authority ends at idx684; all terminal French locator and inherited English files remain unchanged and unpromoted.

²**Source note** [SGA6-XIV-IDX702-XREF96-VS-XREF46-SRCDEF-001]. The printed French note reads “(9.6)”; the contextual reference is formula (4.6).